Venkata Lakshmi Parimala Pasupuleti

Machine Learning Software Engineer · Python · PyTorch · HuggingFace · Transformer Architectures · Multimodal AI · Eval-Driven Development · FastAPI · Distributed Systems

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PROFESSIONAL SUMMARY

Machine Learning Software Engineer with 8+ years of production engineering experience and deep applied ML practice. At GMU, ran the full ML development cycle: data collection, feature engineering, model training (HuggingFace Transformers + PyTorch), evaluation, and inference API deployment - presenting model design, training methodology, and results to faculty. Built AgentProbe: open-source AI evaluation toolkit with CI-integrated regression harness, 5-metric scoring engine, and LLM-as-judge failure analysis - used to evaluate LeaseI, an 8-agent AI platform, achieving 93.6% precision and 87.9% F1 across 1,240 test cases. At Entrata, instrumented production systems with observability tooling (OpenTelemetry, Prometheus, Grafana) and owned distributed backend systems serving 100K+ monthly active users at 99.5% uptime. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models, evaluation pipelines, and data APIs; debugging reliability issues across the application stack.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- **Ran the full ML development cycle end-to-end:** data collection (WHO + UK Road Safety datasets, 190+ countries), feature engineering, model training, evaluation (R^2=0.977), and deployment. Fine-tuned a custom classification model using HuggingFace Transformers + PyTorch (transformer architecture, training loop, quantitative evaluation); deployed inference API on Quart; presented model design, training methodology, and results to faculty.
- Applied Isolation Forest anomaly detection to automate data quality checks across 3 research pipelines — surfacing anomalies that manual review missed. Statistical analysis: hypothesis testing, confidence intervals, cross-validated model performance. Delivered Python programming workshops to 600+ students.

Entrata (Enterprise SaaS / Real Estate Technology) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Performance engineering across production systems** — optimized 200+ SQL queries (30% performance gain) via execution plan analysis, indexing, and partitioning; Docker microservices redesign yielding 50% scalability improvement and 40% deployment time reduction; instrumented systems with OpenTelemetry, Prometheus, Grafana (40% reduction in debugging time).
- **Built fault-tolerant event-driven distributed systems** (RabbitMQ, AWS S3/EC2/RDS) for 100K+ monthly active users at 99.5% uptime — full SDLC ownership across 120+ Agile sprints: design → implement → test → deploy → operate → incident response. Mentored 5+ engineers; 300+ peer code reviews.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built backend modules (Java, Python, PostgreSQL); led AWS S3 migration; data pipelines across hundreds of thousands of records.

PROJECTS

AgentProbe: AI Evaluation Infrastructure EVAL-DRIVEN CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-judge failure analysis; CI-integrated regression harness with golden datasets; distributed model routing across OpenAI/Anthropic/Groq/Ollama. Built to quantify algorithm performance offline before reaching production users.

LeaseI: 8-Agent AI Platform PYTORCH + LANGGRAPH PASS@K HARNESS

Python · LangGraph · FastAPI · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- 8-agent AI platform with durable state management, fault-tolerant HITL escalation, and citation-grounded RAG. Evaluated via pass@k harness (multi-trial sampling + LLM-as-judge + regression gate): 93.6% precision, 87.9% F1, pass@1 81.7%, pass@3 88.3% across 1,240 residential lease PDFs. LLM reasons; deterministic logic makes every regulated decision.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), Java (Spring Boot), JavaScript, TypeScript, SQL, PHP, Bash, R — FastAPI, Django, React.js, REST APIs

ML & Deep Learning: PyTorch, HuggingFace Transformers, Transformer Architectures, TensorFlow, scikit-learn, Isolation Forest, Random Forest, Anomaly Detection, Multimodal AI, Feature Engineering, Model Training, Failure Analysis, Performance Profiling, Model Deployment, Model Versioning, Confidence Intervals, Hypothesis Testing, NumPy, Pandas, Jupyter, Tableau

AI Evaluation: LLM Evaluation, LLM-as-Judge, Eval-Driven Development, Golden Datasets, Regression Harnesses, Pass@k, Offline + Online Evaluation, Scoring Methodology, Ground Truth, AgentProbe, LangChain, LangGraph, RAG Pipelines, ChromaDB

Cloud & Infra: AWS (S3, EC2, RDS, Lambda, Bedrock, SageMaker), Docker, Kubernetes, CI/CD (GitHub Actions), Terraform, OpenTelemetry, Prometheus, Grafana, Distributed Systems, Microservices, PostgreSQL, Redis